

ECON 381: Homework 2

Fall 2026

Due: Thursday, September 17, 2026, at 11:59 p.m. Eastern Time

Submit electronically through the Canvas **Assignments** section.

Instructions and submission guidelines

1. This assignment contains only the analytical portion: **five conceptual questions, five true-or-false questions, and two analytical questions**. The assignment is worth **100 points**: 25 points for conceptual questions, 15 points for true-or-false questions, and 60 points for analytical questions.
2. You may submit individually or in a group of two, as allowed by the syllabus. A pair should make **one group submission**, list both names at the top of the submitted file, and follow the submission instructions given in this document. Group members receive the same grade.
3. Submit your written work as a **single PDF file**. Only **handwritten** solutions are acceptable; they may be written on paper and scanned or written on a tablet and exported as a PDF. The submission must be **clear and legible**. Photos taken by phone or camera are often difficult to read and may be treated as illegible. **Illegible submissions will receive no credit**. Please make sure that **all pages and answers are clearly reproduced and included in your homework PDF**. The file uploaded by the deadline will be treated as your final submission, and you are responsible for its completeness and legibility.
4. Use the following filename format:

Individual: ECON381_HW2_LastName.pdf
Pair: ECON381_HW2_LastName1_LastName2.pdf

5. Late work is accepted according to the syllabus policy. The percentage below is applied to the score the submission otherwise earns:

Time submitted after the deadline	Percentage of earned score retained
At most 3 hours	80%
More than 3 and at most 6 hours	60%
More than 6 and at most 9 hours	40%
More than 9 and at most 12 hours	20%
More than 12 hours	0%

Plan ahead when you anticipate a conflict; illness, university events, and other documented circumstances do not ordinarily excuse a late submission.

6. Any use of generative AI must follow the syllabus disclosure and attribution policy. If you use such a tool, include the required acknowledgment naming the tool and provider and explaining the purpose for which it was used.

I. Conceptual Questions (25 points)

Answer each question carefully. State any assumptions needed for your claims.

1. Let $Y_1, \dots, Y_n$ be a random sample from a population with mean $\mu_Y = E[Y]$. [5 points]
 - (a) State the **Law of Large Numbers (LLN)** for the sample average $\bar{Y}$, including its assumptions and a mathematical statement. [2 points]
 - (b) State the **Central Limit Theorem (CLT)** for the sample average, including its assumptions and a mathematical statement. [2 points]
 - (c) In one or two sentences, explain the different roles played by the LLN and the CLT. [1 point]
2. Researchers eventually observe one sample and obtain one numerical estimate. Why, then, do they care about **sampling error**? Equivalently, why do researchers treat an estimator such as $\bar{Y}$ or $\hat{\beta}_1$ as a random variable even though the reported estimate is a single number? (Hint: Recall the repeated-sampling thought experiment in class.) [5 points]
3. A researcher has an estimate $\bar{Y}$ and its standard error $SE(\bar{Y})$. [5 points]
 - (a) Show how to construct an approximate two-sided **99% confidence interval** by adjusting the critical value. State the appropriate standard-normal critical value and explain where it comes from. [2 points]
 - (b) Holding the population dispersion and confidence level fixed, what happens to the standard error and the width of the confidence interval as the sample size becomes larger? [2 points]
 - (c) Does a larger sample guarantee that the one realized interval contains the true mean μ_Y ? Briefly explain. [1 point]
4. Distinguish the **sample standard deviation of Y** from the **standard error of $\bar{Y}$** . What do they measure in each case, and how are the two quantities related? [5 points]
5. If we observe a very large and positive covariance between X and Y , does that imply the correlation between X and Y is close to one? Explain your answer. [5 points]

II. True or False (15 points)

For each statement, write **True** or **False** and justify your answer in one to three sentences. Credit is based on the explanation, not only on the label.

1. A 95%-level confidence interval means that, with repeated samples, the parameter may take different values, and it will fall into the fixed confidence interval with probability 95%. **[3 points]**
2. A standard error measures the variability of the variable of interest in the population. **[3 points]**
3. If the population correlation between X and Y equals 1, then the realized observations of X and Y should all lie on a straight line with a positive slope. **[3 points]**
4. The Law of Large Numbers implies that whenever the sample size is increased, the new sample average *must be* closer to the population mean than the old sample average. **[3 points]**
5. For the same estimate and standard error, a two-sided 99% confidence interval is *narrower* than a two-sided 95% confidence interval because it leaves less probability in the two tails of the standard normal distribution. **[3 points]**

III. Analytical Questions (60 points)

Show the formulas you use and enough intermediate work to make your calculations clear.

1. **A sample mean and its uncertainty. [30 points]** A researcher observes the following five values of a variable Y :

$$8, \quad 10, \quad 12, \quad 14, \quad 16.$$

Treat these observations as an i.i.d. random sample from a population with unknown mean μ_Y and finite variance.

- (a) Compute the sample average $\bar{Y}$. **[3 points]**

- (b) Compute the sample variance

$$s_Y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

and the sample standard deviation s_Y . **[5 points]**

- (c) Compute the estimated standard error $\text{SE}(\bar{Y}) = s_Y/\sqrt{n}$. **[4 points]**

- (d) For this exercise, use the normal critical value 1.96 to construct the approximate 95% confidence interval

$$\bar{Y} \pm 1.96 \text{SE}(\bar{Y}).$$

Report both endpoints to two decimal places. **[5 points]**

- (e) With only $n = 5$ observations, would you conclude that this realized interval will contain the true mean μ_Y ? Discuss both (i) what 95% confidence level means for this particular interval and (ii) whether the large-sample normal approximation is convincing here. **[6 points]**

- (f) Now suppose a new i.i.d. sample of $n = 1000$ happens to have the same sample average and sample standard deviation you calculated above. Compute its standard error and approximate 95% confidence interval. Compare its width with the interval for $n = 5$. Explain your finding. **[7 points]**

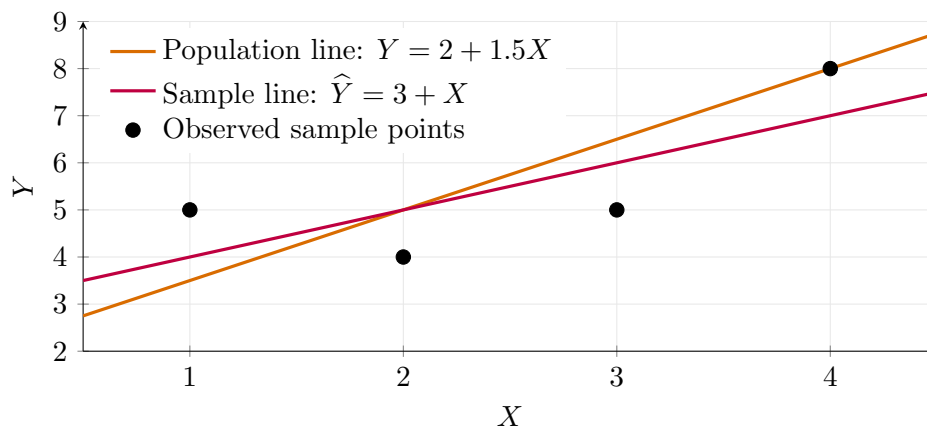
2. **Population and sample regression lines. [30 points]** Suppose the population regression model is

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad \beta_0 = 2, \quad \beta_1 = 1.5.$$

From one sample, researchers estimate the sample regression line

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i, \quad \hat{\beta}_0 = 3, \quad \hat{\beta}_1 = 1.0.$$

The population and sample lines, together with four observed sample points, are plotted below.



- (a) Which intercept and slope coefficients are **population coefficients**, and which are **sample estimates**? Which quantities are fixed across repeated samples, and which generally change from sample to sample? **[6 points]**
- (b) For each observation, calculate

$$\text{population-line value} = \beta_0 + \beta_1 X_i, \quad u_i = Y_i - (\beta_0 + \beta_1 X_i),$$

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i, \quad \hat{u}_i = Y_i - \hat{Y}_i,$$

and complete the table. **[12 points]**

i	X_i	Y_i	$\beta_0 + \beta_1 X_i$	u_i	$\hat{Y}_i$	$\hat{u}_i$
1	1	5				
2	2	4				
3	3	5				
4	4	8				

- (c) Using the graph and your table, explain the difference between a regression error u_i and a residual $\hat{u}_i$. Which one is generally computable by the researcher, and why? **[6 points]**
- (d) Verify numerically that the OLS residuals in your table satisfy **[6 points]**

$$\sum_{i=1}^4 \hat{u}_i = 0 \quad \text{and} \quad \sum_{i=1}^4 X_i \hat{u}_i = 0.$$